MTP F26 · TEAM C · RADIO TEAM → ELECTRONICS TEAM

Suggested hardware for the transceiver

From Radio Team (Santiago, Andrian, Marcos) **To** Carlos, Michele, Bikash, Guillem

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New in v0.5 (18 Oct): the micro is now a Raspberry Pi 3 (the team's request; the 3 A+ is suggested). The radio side does not change. On the board:

- The carrier becomes a **HAT-style board** on the Pi's 40-pin header: SP4, radio LDO, 5 V boost, fuel gauge, charger, switches and LEDs.
- A **5.1 V boost rated for at least 2.5 A** now feeds the Pi all the time. It replaces the small boost and switch that only powered the stick.
- The stick uses the Pi's **own USB port**, so PIO-USB, the stick switch and the USBLC6 are gone.
- The Pi has no ADC, so a **MAX17048 fuel gauge** reads the cell instead of ADC3 and the ideal diode.
- The Pi's **WiFi/BT chip** is disabled in software or desoldered. That is our working assumption; Atenea Q22 asks.
- Cost goes to about **70–90 € per box**. The full analysis is on the Team C Pi 3 Switch page.

v0.5.1 (18 Oct, later): the block diagram shows the HAT boundary, the microSD and the GPIO numbers; the pin map adds the mode switch's second GPIO (16), the fuel-gauge alert (25) and the charger status (26).

Earlier: v0.4.4 added the TX/RX role switch and the secured U.FL plug. v0.4.3 added low-battery countermeasures. v0.4.2 moved SRI/MRM to 20 dBm hopping on the same hardware. v0.4 chose one Ebyte E01-ML01SP4 and one flex omni per box.

The radio baseline is 2.4 GHz with the nRF24L01+. Under our split, the electronics team picks the micro and designs the board. This page is the radio team's **suggestion**, built around every hardware rule in the competition PDF, to save you time. It is not a decision. Please challenge it.

In short

1. **Micro module: Raspberry Pi 3, Model A+ suggested** (Model B works if we already own them). It runs Linux on a microSD card and has native USB host ports. Its WiFi/BT chip is **disabled in software or desoldered**, which is our working assumption until Atenea Q22 is answered.
2. **Radio: one Ebyte E01-ML01SP4 per box**, unchanged (genuine nRF24L01+ with PA and LNA; SMD 14.85 × 18 mm, 10 pads at 1.27 mm; IPEX antenna socket). **VCC is 2.0–3.6 V only**, so it keeps its own 3.3 V regulator from the cell, never the Pi's 3V3 pin. The same module goes from the bench to the final boxes.
3. **Board: a 2-layer HAT-style carrier** on the Pi's 40-pin header. It carries the SP4, the radio LDO, the 5.1 V boost that feeds the Pi, the fuel gauge, the charger, ESD protection, and the switch, button and LEDs. The antenna is a Molex flex omni stuck on a box wall, its cable plugged into the radio module: no RF circuitry on our board.
4. **Power: one protected 18650 Li-ion cell** (3000 mAh or more), charged on board through a charge-only USB-C port. The Pi and radio draw about 2 W, so one cell lasts about 4–5 h with the A+. Use two cells in parallel if the boxes stay on longer.
5. **Estimated cost: about 70–90 € per device**, against the 150 € cap (40–55 € with the Pico 2).

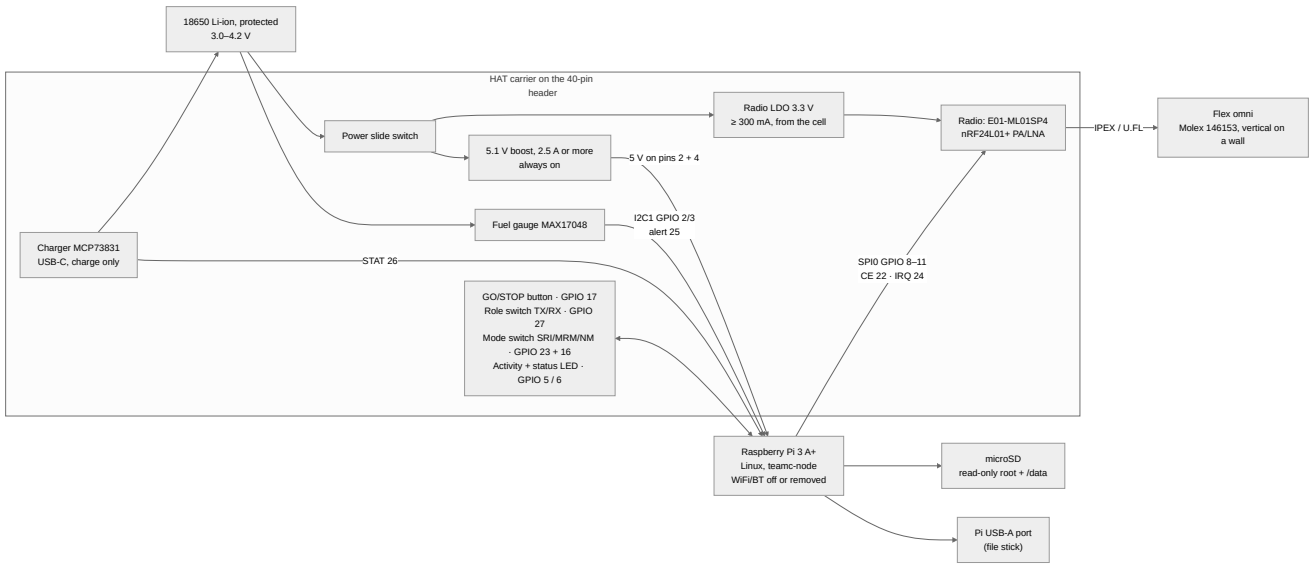
The rules this design answers

RULE (COMPETITION PDF)	WHAT IT FORCES ON THE BOARD
One and only one module including a microprocessor (the rules name "Raspberry..." as an example); no single commercial module embedding the entire system	The Pi 3 is the only micro. The radio, charger, boost and fuel gauge contain no CPU.
No chip implementing WiFi, Bluetooth, ZigBee, 2G–5G	The Pi 3 carries one (CYW43455 on the A+/B+, BCM43438 on the B). **Working assumption:** disabled for good (`dtoverlay=disable-wifi`, `disable-bt`, `rfkill`) or desoldered; **Atenea Q22** asks which one the professors require. Still banned: Pico W / Pico 2 W, ESP32 boards, Arduino Nano 33 BLE/IoT, Uno R4 WiFi, Nano ESP32.

RULE (COMPETITION PDF)	WHAT IT FORCES ON THE BOARD
Read/write the file from/to a USB memory device: 5 min setup while still wrapped in foil, 3 min to upload afterwards	A Pi USB-A port at a box wall (or a short internal extension to a panel socket). The microSD card holds the file, so storage is no longer a constraint. The stick can be removed during the round.
Fits entirely in the 15 × 15 × 7 cm IKEA 365+; only connectors, switches and buttons may stick out	Pi 3 A+ 65 × 56 mm (B: 85 × 56 mm), about 30 mm tall with the HAT. The antenna sits inside: a 35 × 9 mm flex omni, vertical, as high as possible on a wall.
Operates autonomously	A battery and a power switch. Nothing plugged in during a round.
At least one Tx/Rx switch or button; a pilot light that blinks in Tx/Rx	Panel pushbutton for GO/STOP, a **TX/RX role switch** (the professor picks the sender after loading; Atenea Q8), and an activity LED. The box is transparent, so the LED can stay inside.
Extra switches allowed (e.g. network mode), but **no switch indicating the position in the network**	A 3-position mode switch SRI / MRM / NM (two GPIOs). Nothing that encodes a node number.
Drop test from table height to concrete (before MRM)	Pi on M2.5 standoffs; HAT soldered to the header pins or held by four standoffs; **microSD taped or glued** (the Pi 3 slot holds it by friction only); battery strapped; antenna glued down; **U.FL plug secured**.
Designed to self-recover from interruptions	Linux boots in 10–30 s instead of the Pico's 0.3 s. Target ≤ 10 s, hardware watchdog through systemd, read-only root so a power cut cannot corrupt the card. Sim T19: one 30 s reboot in MRM still delivers the file in 90.8 s.
ESD and RF exposure compliance	TVS/ESD protection on the buttons and the 5 V input. RF power is small (≤ 10 mW e.i.r.p.; ≤ 100 mW e.i.r.p. while hopping in SRI/MRM).
Modules shown at Quick Mode are final	The QM prototype uses the same Pi 3 model and the same E01-ML01SP4 as the final board, and the same flex antenna. Q22 must be answered before 6 Nov.
Unit cost ≤ €150; lab-made PCB counts €15; everything (screws, foam, wires, owned boards) goes in the BoM	A Pi we already own still counts at market price.

Block diagram

The only fast data path is SPI to the radio, at 31 kB/s (250 kbps). USB file transfer happens only before and after a round.



The battery, the antenna and the Pi's own parts (microSD, USB port) sit outside the HAT. The software that runs on this hardware, `teamc-node`, is written and tested against an emulated radio; it has not run on a Pi yet (Team C Protocol Proposal v0.4.1).

Micro module options

The team wants a Pi 3. The table ranks the Pi 3 variants and keeps the alternatives for comparison.

MODULE	WIRELESS CHIP	DRAW AT 5 V	PRICE	VERDICT	NOTES
Raspberry Pi 3 Model A+	CYW43455, off or removed	`~0.30–0.40 A`	~€27	suggested	Lowest draw and smallest Pi 3 (65 × 56 mm); one USB port wired straight to the SoC; no Ethernet or hub chip to power. Built until at least Jan 2030.

MODULE	WIRELESS CHIP	DRAW AT 5 V	PRICE	VERDICT	NOTES
Raspberry Pi 3 Model B	BCM43438, off or removed	~0.35–0.45 A	~€37	if owned	85 × 56 mm, 4 USB ports behind a LAN9514 hub. Fine if the team already has them.
Raspberry Pi 3 Model B+	CYW43455, off or removed	~0.45–0.60 A	~€40	highest draw	Shortest battery life of the three.
Compute Module 3+ (Lite)	None	~0.3 A	~€27–33	fallback	Same processor with no wireless chip at all, if Atenea says the chip must go and desoldering fails. Needs a SODIMM carrier we would design. Stock is patchy.
Raspberry Pi Zero (non-W)	None	~0.1–0.2 A	~€5–10	alternative	Linux too, no wireless question, a third of the draw. Single core, so real-time timing is harder; the USB host needs an OTG adapter.
Raspberry Pi Pico 2 (RP2350)	None	~0.03–0.05 A	~€5	previous plan	Cheapest, 0.3 s reboot, exact timing. Its port (PIO-USB, flash rules) stays in git history.
Pico W / Pico 2 W, ESP32, Nano 33 BLE/IoT, Uno R4 WiFi, Pi Zero W, Pi 4/5	—	—	—	banned	On-board WiFi/BT chip, and nothing to gain over the Pi 3.
Micro + radio combos (Feather	—	—	—	avoid	Risks the "single commercial module

MODULE	WIRELESS CHIP	DRAW AT 5 V	PRICE	VERDICT	NOTES
RP2040 RFM69, Moteino)					embedding the entire system" rule.

Prices are estimates seen online in October 2026. Get real quotes before ordering.

Suggested parts per device

Example part numbers to start from. Electronics owns the final choice, the footprints and the sourcing.

FUNCTION	SUGGESTED PART	WHY	≈ € / DEVICE
Micro module	Raspberry Pi 3 Model A+ (B if owned)	See above. Owned boards count at market price	27–37
Storage	microSD 16 GB, high-endurance or industrial	Holds Linux and the files. Read-only root, small data partition	5–8
Radio module	1 × Ebyte E01-ML01SP4 (nRF24L01P + PA/LNA, SMD, IPEX). **Not** E01C-* (clone chip)	Chip-controlled PA/LNA, so only SPI + CSN + CE + IRQ. Same module from the bench (on a breakout) to the final box	≈ 4–6
Antenna	Molex 146153-0100 flex omni (34.9 × 9 mm, 100 mm cable, U.FL)	Plugs straight into the module. Vertical polarisation, mounted high on a wall	≈ 3
5 V for the Pi	Boost 5.1 V, ≥ 2.5 A out from 3.0–4.2 V (TPS61088-class + inductor; a ready-made 5 V 3 A module for the bench)	The Pi needs a steady 5 V; it flags undervoltage below 4.63 V. Near the 3.0 V cutoff the boost pulls about 2 A from the cell	3–5
Battery sense	MAX17048 fuel gauge on I2C1	The Pi has no ADC. Reads cell voltage and state of charge	2
Protection	TVS + reverse-polarity protection on the 5 V feed; TVS on the button lines	Feeding the header pins bypasses the Pi's own input protection	0.6
Battery	1 × protected 18650 Li-ion (3000 mAh or more) with solder tabs, or	Steel can is robust in the drop test; rechargeable (sustainability report)	6–8

FUNCTION	SUGGESTED PART	WHY	≈ € / DEVICE
	holder + strap; a second cell in parallel if needed		
Charger	MCP73831 at 500 mA + USB-C receptacle, charge only	Charges a 3000 mAh cell overnight (6–7 h). Nothing is plugged in during operation	1.2
Controls	Slide power switch, panel GO/STOP pushbutton, TX/RX role switch, 3-position mode switch (on-off-on), 2 LEDs	Required by the rules	2.5
Radio supply	3.3 V LDO (AP2112K-3.3 or TLV75533, dropout ≤ 0.1 V at 150 mA) from the cell + 100 µF / 10 µF / 100 nF at the module	SP4 VCC is 2.0–3.6 V (abs. max 3.6 V); PA/LNA modules are sensitive to supply noise; handles the ~120 mA TX bursts. Never the Pi's 3V3 pin	1
PCB	2-layer HAT-style carrier, 65 × 56 mm (A+ outline), 40-pin header	ETSETB lab = €15 in the BoM (rule); an external fab is ~€1 per board but adds 1–2 weeks	1–15
Mechanics	M2.5 standoffs (58 × 49 mm pattern), base plate, cable ties, hot glue, tape for the SD card, IKEA 365+ box	Drop test; the BoM must list them too	5–7
Total per device (verify with quotes)			**≈ 70–90**

Power budget

Current draw

Pi 3 A+ at 5 V	≈ 0.30–0.40 A, one core busy-waiting for the radio ESTIMATE
SP4 TX	≈ 120 mA peak DATASHEET
SP4 RX	≈ 26 mA DATASHEET
USB stick (Pi port)	50–100 mA, only during load/save

Battery sizing

Usable energy of one 3000 mAh cell is about 3.0 Ah × 3.6 V × 85 % boost efficiency ≈ 9.2 Wh. With the radio's ~0.15 W that gives:

- Pi 3 A+, 1.7–2.2 W: **4.2–5.4 h**
- Pi 3 B, 2.0–2.4 W: 3.8–4.6 h
- Pi 3 B+, 2.4–3.2 W: 2.9–3.8 h

A competition day of ~4 h powered on fits the A+ on one cell, barely. Use two cells in

	ESTIMATE
From the cell	≈ 0.5–0.8 A; up to ≈ 2 A at boot near 3.0 V

parallel, or switch the box off between rounds (boot ≤ 10 s). Measure the real draw at T0.

Low battery

The radio LDO from the cell is still fine: a clean 3.3 V down to a cell of about 3.4 V, about 2.95 V at the 3.0 V cutoff, inside the SP4's 2.0–3.6 V. The new weak point is the **5 V boost**: if it sags below 4.63 V the Pi throttles, and lower it crashes.

- **Battery sense:** MAX17048 on I2C1 (no ideal diode needed any more). The firmware also reads the Pi's undervoltage flag (`vcgencmd get_throttled`) and logs it.
- **Firmware:** thresholds (3.5 / 3.4 / 3.2 V) re-measured at T0 under the Pi's load, since the cell sags more at 0.5–1 A. Below the lowest, fsync received data after every burst.
- **T0** measures the radio output and the Pi's current at 3.6, 3.3 and 3.0 V.
- **Boost:** size the inductor and switch for 2 A input at 3.0 V; 100 µF or more at the Pi's 5 V pins.
- **Radio fallback** (unchanged): a buck-boost (TPS63001/TPS63020) if T0 shows more than 2 dB less output at 3.0 V.

For the sustainability report

The Pi 3 box uses about 7× the energy of the Pico design (≈ 2 W against ≈ 0.3 W). In its favour: the Pi is reflashable, repairable and reusable after the course, and the HAT unplugs. We can measure watts per round directly with the fuel gauge.

Build stages

STAGE	BOARD	GOAL	TARGET
1 · Bench	Pi 3 + E01-ML01SP4 on a small breakout (1.27 → 2.54 mm) wired to the header,	Radio tests T0 (output power, Pi noise in our band) and T2 (frame gap on Linux); USB-	as soon as parts arrive

STAGE	BOARD	GOAL	TARGET
	radio LDO from a bench supply, Pi on its own 5 V supply	stick read/write on the Pi's port	
2 · Quick Mode	Perfboard HAT, **same Pi model and module part numbers**, external power allowed	One error-free line from a file in < 2 min, first attempt	week 6–7
3 · PCB rev A	HAT-style carrier (KiCad), boxed, battery	Autonomous, drop-tested, range tests T4/T5	week 8–9
4 · PCB rev B	Fixes from rev A	Pretest (4 Dec) fully packaged	week 10–11

Risks and how we cover them

RISK	MITIGATION
The professors require the WiFi/BT chip to be removed	Atenea Q22 before Quick Mode. Try desoldering on one spare board now. Last resort: Compute Module 3+ (no chip).
SD card corrupted by a power cut or the drop test	Read-only root (overlayfs) with a small data partition, high-endurance card, card taped or glued in its slot.
Linux reboot after an "attack" or a brown-out takes 10–30 s	Boot ≤ 10 s target, hardware watchdog via systemd, resume from the last acknowledged line (sim T19: 90.8 s in MRM with a 30 s reboot).
The 5 V boost sags at low charge and the Pi crashes	Boost sized for 2 A input at 3.0 V, bulk capacitance, fuel gauge warning, a second cell in parallel if T0 shows trouble.
Pi clocks (6 × 400 MHz = 2400 MHz, HDMI) add noise in our band	HDMI off, SP4 at the board edge away from the SoC, antenna high on the wall; T0 logs RPD hits per hop channel with the Pi idle and busy, and drops a bad channel.
The foil-wrapped setup makes the USB port hard to reach	A Pi USB port facing a side wall, or a short internal extension to a panel socket. Atenea Q6 asks whether the foil may be opened at the port.
Clone radio chips (Si24R1)	Buy only the plain E01-ML01SP4 part number from Ebyte's store or a distributor; T0/T2 check the chip.

RISK	MITIGATION
SMD module needs a breakout before any PCB exists	A one-evening lab board (or careful wiring) in stage 1; the HAT takes over in stage 3.
Drop test pulls the HAT off the header or ejects the SD card or battery	HAT soldered or on four standoffs; SD taped; strapped cell; drop rev A on purpose in week 9.

What we need from electronics

1. Confirm the Pi 3 model (A+ suggested) and how many boards we already have, by **Fri 23 Oct**. The radio order (SP4s, antennas, pigtails) goes out now regardless; the Pico 2s are dropped from it.
2. Try desoldering the WiFi/BT chip on one spare board, then check that the Pi still boots and that SPI and USB work.
3. Confirm the pin map (proposal: SPI0 on GPIO 9/10/11, CSN = CE0 on GPIO 8, CE GPIO 22, IRQ GPIO 24, I2C1 GPIO 2/3 for the fuel gauge, button GPIO 17, role switch GPIO 27, mode switch GPIO 23 + 16, LEDs GPIO 5/6, fuel-gauge alert GPIO 25, charger status GPIO 26). The full table is on the Pi 3 Switch page.
4. The power tree: 5.1 V boost part, fuel gauge, protection on the 5 V feed. The KiCad brief (`hw/KICAD_SESSION_PROMPT.md`) is updated for the HAT.
5. The IKEA box's inside height and where the antenna can sit highest.

Radio details (interface pins, test plan, 20 dBm hopping in SRI/MRM): Team C Radio Brief v0.11. Firmware and USB: Team C Protocol Proposal v0.4.1. Pi 3 analysis: Team C Pi 3 Switch. Rules quoted from *MTP F26 – Program Competition Rules – Network Upgrade III + Sustainability*.

Sources checked 18 Oct 2026: Pi 3 A+ · Pi 3 B · Pi power and USB documentation · measured Pi 3 current · Ebyte E01-ML01SP4 manual · nRF24L01+ Product Specification.